Observable GR Transport and Source-Side Action Automation

Blind Fisher–Energy Sector Selection, Levi-Civita Placement, and Effective Source Construction

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July 28, 2026

Abstract

This paper combines two conservative roles for Resolution Geometry (RG) in gravitational modelling. First, general relativity is placed in RG not by identifying the Lorentzian spacetime metric with the positive-definite observation metric, but by using the Levi-Civita connection as transport data acting on an observable resolution projector. Second, the source-side object presented to a gravitational equation is not assembled by manually assigning resolved scalars to stress-energy components; it is generated by metric variation of an effective observation action. The missing link between these two statements is the observation layer: given an observation channel B, covariance Σ, and metric-dependent scalar/action Hessian H_g, the resolved/null sector is constructed by the generalized Fisher–energy eigenproblem

$$Fv_i = \lambda_i H_g v_i, \qquad F = B^T \Sigma^{-1} B.$$

The resulting projectors P_R, P_N are then transported by the Levi-Civita connection through

$$A_{RN} = P_N(\nabla^{\mathrm{LC}} P_R)P_R,$$

and used to define the source-side elimination problem

$$W_{\mathrm{obs}}[\phi_R, g; \mathcal{R}] = S[\phi_R, \phi_N^\star; g], \qquad T^{\mathrm{obs}}_{\mu\nu} = -\frac{2}{\sqrt{-g}} \frac{\delta W_{\mathrm{obs}}}{\delta g^{\mu\nu}}.$$

The result is a finite-resolution pipeline

$$(M, g, \nabla^{\mathrm{LC}}) + (B, \Sigma, S) \longrightarrow (F, H_g) \longrightarrow (P_R, P_N) \longrightarrow A_{RN} \longrightarrow W_{\mathrm{obs}} \longrightarrow T^{\mathrm{obs}}_{\mu\nu}.$$

Numerical batteries on flat, FRW-like, Schwarzschild exterior, nonlinear scalar, and degeneracy-controlled systems support the individual gates. The paper states only placement and automation claims; it does not derive GR, does not choose the experiment, and does not reconstruct information lying in the Fisher hole.

Contents

1 Purpose

The problem is to join two previously separate tasks. The first task is *placement*: where does Lorentzian GR data live inside an RG framework that requires a positive-definite observation geometry? The second task is *source automation*: once a finite-resolution sector has been selected, how is an effective source tensor generated without a manual stress dictionary?

The proposed chain is

$$\boxed{(M, g, \nabla^{\mathrm{LC}}) + (B, \Sigma, S[\phi; g]) \longrightarrow (F, H_g) \longrightarrow (P_R, P_N) \longrightarrow A_{RN} = P_N(\nabla^{\mathrm{LC}} P_R)P_R \longrightarrow W_{\mathrm{obs}} \longrightarrow T^{\mathrm{obs}}_{\mu\nu}.}$$

The finite observer supplies an observation channel B and covariance Σ. The metric-dependent field model supplies $S[\phi; g]$ and hence a Hessian H_g. The Fisher–energy geometry supplies the resolved/null split. GR supplies transport through ∇^{LC}. The source tensor is generated by metric variation of W_{obs}.

Non-claim 1.1 (No derivation or replacement of GR)**.** The paper does not derive the Einstein tensor, the Einstein equation, or the Lorentzian metric. The geometric side of GR remains in its usual role.

Non-claim 1.2 (No automatic detector selection)**.** GR alone does not define B or Σ. The measurement model must be supplied. The claim is that, once B, Σ and H_g are supplied, the sector split is automated.

Non-claim 1.3 (No recovery from the Fisher hole)**.** If the Fisher form contains null or near-null directions, they are reported as unresolved. The construction does not reconstruct information absent from the observation channel.

2 Layer I: blind observation geometry

Let $B : \phi \mapsto y$ be a finite observation map and let $\Sigma \succ 0$ be its observation covariance. Define

$$F = B^T \Sigma^{-1} B.$$

Let $S[\phi; g]$ be a metric-dependent scalar/action model, with local quadratic Hessian

$$H_g = \frac{\partial^2 S}{\partial \phi^2}.$$

Definition 2.1 (Fisher–energy sector split)**.** Assume $H_g \succ 0$ and $F \succeq 0$. Solve

$$Fv_i = \lambda_i H_g v_i, \qquad \lambda_1 \geq \lambda_2 \geq \cdots .$$

For a budget k or threshold rule, define

$$R_k = \mathrm{span}\{v_1, \dots, v_k\}, \qquad N_k = R_k^{\perp_{H_g}}.$$

The projectors onto these subspaces are denoted P_R and P_N.

Theorem 2.2 (Optimal Fisher capture)**.** *Among all k-dimensional subspaces with H_g-orthonormal basis Q, the top generalized eigenvectors maximize*

$$\mathrm{tr}(Q^T F Q), \qquad Q^T H_g Q = I_k.$$

Proof. Let $H_g = LL^T$ and set $U = L^T Q$. Then $Q^T H_g Q = I$ iff $U^T U = I$, and

$$\mathrm{tr}(Q^T F Q) = \mathrm{tr}\left(U^T L^{-1} F L^{-T} U\right).$$

By Rayleigh–Ritz, the maximizing U is spanned by the top eigenvectors of $L^{-1}FL^{-T}$. Transforming back gives the generalized eigenvectors of $Fv = \lambda H_g v$. □

Corollary 2.3 (Automatic observation projector)**.** *Given (B, Σ, H_g) and a declared budget or threshold, P_R and P_N are determined by the observation/action geometry. They are not manually chosen coordinate sectors.*

3 Layer II: GR placement as transport

Resolution Geometry uses a positive-definite observation geometry. A Lorentzian GR metric cannot be inserted into that slot without breaking orthogonal resolved/null decomposition. GR therefore enters through its Levi-Civita connection.

Definition 3.1 (GR-induced resolved–null transport block)**.** Given P_R, P_N constructed from (F, H_g), define

$$\boxed{A_{RN} = P_N(\nabla^{\mathrm{LC}} P_R)P_R.}$$

This is the off-diagonal resolved–null component of a connection acting on the resolution projector.

Proposition 3.2 (Layered GR placement)**.** *Given $(M, g, \nabla^{\mathrm{LC}}; B, \Sigma, S[\phi; g])$ with $H_g \succ 0$, the finite-resolution GR placement chain is*

$$(M, g, \nabla^{\mathrm{LC}}; B, \Sigma, S) \longrightarrow (F, H_g) \longrightarrow (P_R, P_N) \longrightarrow A_{RN} = P_N(\nabla^{\mathrm{LC}} P_R)P_R.$$

The Lorentzian metric remains in the GR layer; the RG split is constructed in positive-definite Fisher–energy geometry.

Remark 3.3 (Groupoid level)*.* The block A_{RN} is not itself a groupoid arrow. A compatible connection can be exponentiated or path-ordered to produce transport between local resolution geometries. Closed loops yield automorphisms only if the transported split and covariance close.

4 Layer III: source-side action automation

Once a resolved/null sector is available, source construction proceeds at the action level. Write

$$\phi = (\phi_R, \phi_N).$$

The unresolved field is eliminated at fixed resolved field:

$$\partial_{\phi_N} S[\phi_R, \phi_N^\star; g] = 0.$$

For convex Euclidean sectors this is the minimizer

$$\phi_N^\star = \arg\min_{\phi_N} S[\phi_R, \phi_N; g].$$

Definition 4.1 (Observation action)**.** The classical observation action is

$$\boxed{W_{\text{obs}}[\phi_R, g; \mathcal{R}] = S[\phi_R, \phi_N^\star; g].}$$

Theorem 4.2 (Metric-variation envelope identity)**.** *Assume the eliminated branch $\phi_N^\star(g)$ is differentiable and stationary. For any metric/background parameter λ,*

$$\frac{\partial W_{\text{obs}}}{\partial \lambda} = \left.\frac{\partial S[\phi_R, \phi_N; g]}{\partial \lambda}\right|_{\phi_N = \phi_N^\star}.$$

Proof. By the chain rule,

$$\frac{d}{d\lambda} S[\phi_R, \phi_N^\star(\lambda); g(\lambda)] = \frac{\partial S}{\partial \lambda} + \left(\frac{\partial S}{\partial \phi_N}\right)^T \frac{\partial \phi_N^\star}{\partial \lambda}.$$

The second term vanishes by stationarity of $\phi_N^\star$. □

Corollary 4.3 (Automated observable stress)**.** *The resolved source response is generated by metric variation:*

$$\boxed{T_{\mu\nu}^{\text{obs}} = -\frac{2}{\sqrt{-g}} \frac{\delta W_{\text{obs}}}{\delta g^{\mu\nu}}.}$$

Thus stress components are generated by the effective action rather than by assigning resolved scalars manually to energy density, pressure, flux, or shear.

5 Fisher health and non-identifiability

The Fisher spectrum also supplies a health diagnostic. For eigenvalues λ_i and threshold τ, define

$$r_\tau = \#\{i : \lambda_i/\lambda_1 > \tau\}, \qquad h_\tau = n - r_\tau,$$

with effective rank

$$r_{\text{eff}} = \frac{(\sum_i \lambda_i)^2}{\sum_i \lambda_i^2}.$$

Large h_τ, small r_{eff}, or tiny λ_n/λ_1 indicates that the fit may be accurate in observation space while ambiguous in parameter/source space.

Theorem 5.1 (Null directions cannot be recovered)**.** *For a linear forward map $f(\theta) = J\theta$, any $v \in \ker J$ satisfies*

$$f(\theta + av) = f(\theta)$$

for all scalars a. Therefore no method using only $f(\theta)$ can distinguish θ from $\theta + av$.

Proof. Since $Jv = 0$, $J(\theta + av) = J\theta + aJv = J\theta$. □

6 Numerical evidence

This section summarizes the batteries supporting the combined chain. The detailed outputs are listed in the reproducibility appendix.

6.1 Blind sector selection

The blind sector battery tested flat, random conformal, FRW, and Schwarzschild exterior backgrounds. The same metric-to-action and sector-selection map was used after metric arrays were supplied. On all tested cases, automatic sectors captured essentially all Fisher-visible content at the chosen budget, while random same-dimensional sectors captured about one quarter. Stationarity, envelope, and quotient checks passed at numerical precision.

Background	stationarity rel.	envelope rel.	auto capture	random capture
Flat	4.12×10^{-12}	7.64×10^{-10}	1.000	0.249
Random conformal	1.57×10^{-9}	1.50×10^{-9}	1.000	0.249
FRW	1.03×10^{-11}	1.27×10^{-9}	1.000	0.249
Schwarzschild exterior	6.32×10^{-13}	7.48×10^{-10}	1.000	0.252

6.2 Final GR F-layer integration battery

The final integration battery tested the full finite-dimensional chain

$$g \to H_g, \qquad B \to F, \qquad (F, H_g) \to P_R, P_N, \qquad \nabla^{\mathrm{LC}} P_R \to A_{RN}.$$

It used a 10×10 scalar/action grid, 2×2 block observations, and $k = 25$ resolved directions. Three cases were tested: flat control, a controlled FRW-like transport perturbation, and a Schwarzschild exterior transport block seeded by

$$\beta(8M) = 1 - \sqrt{1 - \frac{2M}{8M}} = 0.1339745962.$$

Case	β	$\|A_{RN}\|_F$	$\|A_{RN}\|_2$	leakage $\|\cdot\|_F$	sector angle
Flat	0	0	0	0	0
FRW-like	0.2350528691	0.2870522849	0.2350528691	0.2851430350	0.2350528691
Schwarzschild $r = 8M$	0.1339745962	0.1636130379	0.1339745962	0.1632586419	0.1339745962

The Fisher health vector remained unchanged before and after the finite transport in the transported frame:

Case	rank before	hole before	eff. rank before	rank after	hole after	eff. rank after
Flat	25	75	19.2168	25	75	19.2168
FRW-like	25	75	18.3764	25	75	18.3764
Schwarzschild	25	75	11.0919	25	75	11.0919

Finding 6.1 (Integrated placement check)**.** The flat control gives zero coupling and zero leakage. The FRW-like and Schwarzschild transport inputs give nonzero resolved–null coupling proportional to the declared transport amplitude. Transport changes the sector orientation but does not erase the Fisher hole.

6.3 Source-side action construction

Direct nonlinear scalar elimination verified that W_{obs} can be constructed by actual unresolved-field elimination, not only by a Schur formula. On tested grids, stationarity and local metric envelope identities held at about 10^{-9} precision, quotient invariance held at displayed precision, and null-coupling reduction gave zero error.

6.4 Degeneracy batteries

Degenerate forward-map batteries showed that ordinary fit metrics can remain excellent while Fisher holes grow. In a radial exterior/interior toy, a 24-dimensional interior mapped to five exterior observables had hole dimension 19; a perturbation changed the interior profile by relative size 2.098 while changing exterior data by only 2.39×10^{-13}. This supports the non-claim that unresolved information is not reconstructed; it is measured as a hole.

7 Combined placement theorem

Theorem 7.1 (Observable GR transport and source automation)**.** *Let $(M, g, \nabla^{\mathrm{LC}})$ be a GR background, let B, Σ define a finite observation channel, and let $S[\phi; g]$ be a metric-dependent scalar/action model with $H_g \succ 0$. Suppose an admissible resolved budget or threshold is fixed and an eliminated unresolved branch exists. Then the following objects are generated:*

$$F = B^T \Sigma^{-1} B, \qquad (P_R, P_N) \text{ from } Fv = \lambda H_g v,$$

$$A_{RN} = P_N(\nabla^{\mathrm{LC}} P_R) P_R, \qquad W_{\mathrm{obs}}[\phi_R, g] = S[\phi_R, \phi_N^\star; g],$$

$$T_{\mu\nu}^{\mathrm{obs}} = -\frac{2}{\sqrt{-g}} \frac{\delta W_{\mathrm{obs}}}{\delta g^{\mu\nu}}.$$

The construction places GR transport on an automatically selected finite-resolution sector and generates the observable source by metric variation of the effective action.

Proof. The Fisher–energy generalized eigenproblem constructs P_R, P_N by the optimal Fisher-capture theorem. The Levi-Civita connection differentiates the generated projector, and projection of $\nabla^{\mathrm{LC}} P_R$ to $N \leftarrow R$ gives A_{RN}. The unresolved-field stationarity condition defines $\phi_N^\star$ and hence W_{obs}. The metric variation of the scalar functional yields the observable source tensor. Each step is conditional on its stated admissibility assumptions; no step requires identifying g_{GR} with a positive-definite observation metric. □

8 Conclusion

Paper B and Paper C can be unified. The unified result is not merely a source-side construction and not merely a GR-placement note. It is a finite-resolution pipeline:

$$\boxed{(M, g, \nabla^{\mathrm{LC}}) + (B, \Sigma, S) \to (F, H_g) \to (P_R, P_N) \to A_{RN} \to W_{\mathrm{obs}} \to T_{\mu\nu}^{\mathrm{obs}}.}$$

The F-layer is no longer a manually selected split; it is generated from the observation channel and action geometry. GR enters through Levi-Civita transport of this generated projector. The effective source is generated variationally. The unresolved sector remains visible as the Fisher hole rather than being hidden inside a confident but non-identifiable fit.

A Reproducibility

The numerical outputs used by this paper are saved in the following files:

- rg_blind_fiber_sector_automation_results.txt;
- rg_gr_flayer_integration_battery_results.json;
- rg_gr_flayer_integration_battery_summary.txt;
- rg_actual_phiN_elimination_attack.py;
- rg_degeneracy_tightening_battery_results.json;
- rg_degeneracy_tightening_battery_summary.txt.

The final integration battery is deterministic. It uses 10×10 grids, 2×2 block-average observations, mass parameter $m^2 = 0.7$, resolved budget $k = 25$, Schwarzschild loop radius $r = 8M$, and finite transport block

$$\Omega = \begin{pmatrix} 0 & -B^T \\ B & 0 \end{pmatrix}, \qquad B_{ii} = \frac{\beta}{i+1}, \quad i = 0, \ldots, 5.$$

The finite transport is $U = \exp(\Omega)$. The blind sector and degeneracy batteries use deterministic random seeds as declared in their result files.

References

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